CODERSINPUNE 2026 · ROUND 2: THE BUILD

ReV — Handover Engine

Evidence-Backed Knowledge Transfer & Developer Offboarding Platform

Team:	**ReV Team**
Live Link:	https://rev.fev06.xyz
Source Code:	github.com/FeV-06/ReV (Public)
Model Provider:	**OmniRoute / gemini-3.8-flash-high**

1. Executive Summary & Core Mission

When an experienced engineer leaves a software team, they take weeks of undocumented architectural decisions, recovery procedures, and edge-case warnings with them. Traditional handovers fail because static markdown docs become stale within days, and free-form chats suffer from severe hallucinations.

ReV (Relaying to Everyone Virtually) solves this crisis by transforming fragmented commit histories, Tree-Sitter AST code symbols, and Slack discussions into an **evidence-linked, human-verified developer handover**. ReV enforces mathematical citation integrity: every claim must cite an exact substring from an authorized source artifact.

2. System Architecture: The Three-Tier Local-First Model

ReV is engineered as a hybrid local-first system that decouples native CPU-bound code analysis from client presentation and persistent team synchronization:

1. Headless Daemon `:3078`

Node 22+ background service bound to loopback. Owns persistent PGlite relational storage, orchestrates Git watchers, and executes native Rust core tasks.

2. Web Dashboard `:3077`

Next.js 16 / React 19 UI following Google's **Material 3 Expressive** tokens. Features a 4-stage screen-advancing Studio and Successor Reader.

3. VS Code Extension IDE NATIVE

Dual-bundle extension (Node host + Browser IIFE) embedding the shared React Team Sync panel and workspace-scoped telemetry inside the developer's editor.

Native Rust Core Engine (crates/rev-core) HIGH PERFORMANCE

Compiled native binary that executes Tree-Sitter AST symbol parsing (TypeScript, Rust, and Python fallback), git blame ownership and churn heuristics (`analyze_local_risk`), and verifies verbatim citation quotes before generating candidates.

3. Completed Round 2 Feature Deliverables

- ✔ **4-Stage Material 3 Creation Wizard:** Sequential screen-advancing studio (Workspace & Contributor ➔ Scope Confirmation ➔ Synthesis & Interview ➔ Export & Sovereignty).
- ✔ **Tree-Sitter AST Code Symbol Mapping:** Extracts changed functions, classes, and types from diffs, linking AST symbols directly to Git commit authors.
- ✔ **Grounded Departure Interview Engine:** Core generates standalone gap questions by cross-referencing commit history with repository documentation.
- ✔ **Interactive Successor Q&A Assistant:** Answers successor questions by citing exact verbatim excerpts from codebase artifacts and interview answers.
- ✔ **Priority Day-1 Onboarding Checklist:** Successor portal task tracker for verified codebase setup, credentials, and deployment runbooks with local state.
- ✔ **Local Data Sovereignty Controls:** Dynamic toggle allowing teams to keep handovers strictly local in PGlite or sync to remote Supabase.
- ✔ **Multi-Source Scoping & Granular Invalidation:** Real-time selection of commits and Slack messages; scope edits atomically invalidate prior approvals.
- ✔ **One-Click Cited Markdown Export:** Generates publication-ready `HANDOVER.md` complete with verifiable source citations and next steps.

4. What Changed Since Round 1 (And Why)

Rather than treating Round 1 as an unchangeable specification, we evolved ReV based on realistic operational challenges encountered during early testing:

Pillar / Area	Round 1 Proposal	Shipped in Round 2 (The Build)
System Architecture	Single monolithic web form running purely in-browser against local files.	**Decoupled Three-Tier Architecture:** Headless Desktop Daemon (:3078), Next.js M3 Web Studio (:3077), and native VS Code Extension.
Database & Sync	Local-only single-session storage; no sharing between team members.	**Hybrid PGlite + Supabase Sync:** Local-first PGlite relational storage augmented with Supabase cloud sync and an explicit sovereignty toggle.
Handover Generation	Static LLM text reconstruction without active questioning or follow-ups.	**Departure Interviews & Cited Q&A:** Proactive gap generator asking departing engineers tough architectural questions, plus successor assistant.
User Experience	Stacked single-page forms with developer persona buttons cluttering views.	**Material 3 Expressive System:** Clean role-tailored portals (Admin Studio vs. Successor Onboarding Reader) with animated slide transitions.
Code Intelligence	Basic commit message and diff grep regex matching.	**Tree-Sitter AST Analysis:** Multi-language AST parsing extracting exact functions, structs, and methods directly impacted by commit churn.

5. Transparent Operational Disclosures ("What Does NOT Work Yet")

Transparency Notice: ReV is an evidence-backed product. Below is the honest, unvarnished status of work-in-progress boundaries:

- **Multi-Tenant Route Authorization Gating:** While the isolated Supabase RLS and SQL membership suites pass 38 unit tests, legacy bulk sync is permanently disabled (HTTP 403) and team sync network calls fail closed until per-user route authorization is completely wired across every endpoint.
- **Live Slack OAuth Configuration:** Connecting a live Slack workspace requires pre-configured OAuth bot client credentials. For zero-config evaluation, ReV provides full synthetic ZIP archive ingestion.
- **Unauthenticated GitHub API Bounds:** Without a configured GitHub personal access token (`REV_GITHUB_TOKEN`), public repo imports are constrained by GitHub's 60 requests/hour IP rate limit.
- **Deep Reasoning Latency:** Synthesizing comprehensive handovers with deep reasoning models (e.g. `gemini-3.8-flash-high`) takes 45–90 seconds over large symbol trees. The UI provides native progress indicators, but instant evaluation is accessible via the synthetic sample.

6. Binding Commitments for Round 3 (The Finish)

We commit to delivering the following three substantial, technically rigorous improvements for Round 3:

Commitment 1: Laya Classifier — AI PIPELINE

Commit & AST Semantic Triage: Integrate a specialized classifier pipeline to evaluate whether an AST symbol or commit represents an intentional architectural decision

Commitment 2: Team Sync — MULTI-TENANT

Frictionless Supabase Team Sync: Complete the end-to-end authenticated team mode with verified member sessions, private-by-default owned runs, and frictionless

Commitment 3: Risk Heatmap — RISK ENGINE

Bus-Factor & Churn Heatmap: Deliver an interactive visual risk heatmap identifying repository modules with critical knowledge concentration (combining Git blame

versus routine cosmetic churn, pruning noise before LLM synthesis.

one-click lead sharing across the web dashboard and VS Code extension.

ownership, commit churn, and test coverage gaps via native Rust heuristics).

Judges Verification Guide: Open **https://rev.fev06.xyz** in any modern browser. You can click *"Try sample project"* to run through the complete 4-stage wizard, inspect AST symbol citations, and run the departure interview with zero setup, or provide any public GitHub repo to experience live model inference.